

Diagonality

The Best of Both Worlds

The tactical blog Spielverlagerung argues the diagonal is football's optimum:
the progression of a vertical pass with the safety of a horizontal one.

We put that theory to the test on 3D skeleton data,
and turn it into a live map of where to attack.

Jaime Oriol

DIAGONALITY: THE ORIENTATION-AWARE PIPELINE

FROM 3D SKELETON TRACKING TO A REAL-TIME MAP OF DIAGONAL OPPORTUNITY

→ AWS WORLD SPORTS INNOVATION CUP 2026 • CHALLENGE 2 ←

1 • 3D SKELETON

We track 21 points on every player's body, 50 times a second.

That tells us exactly where each player is looking, and which way his shoulders are facing.



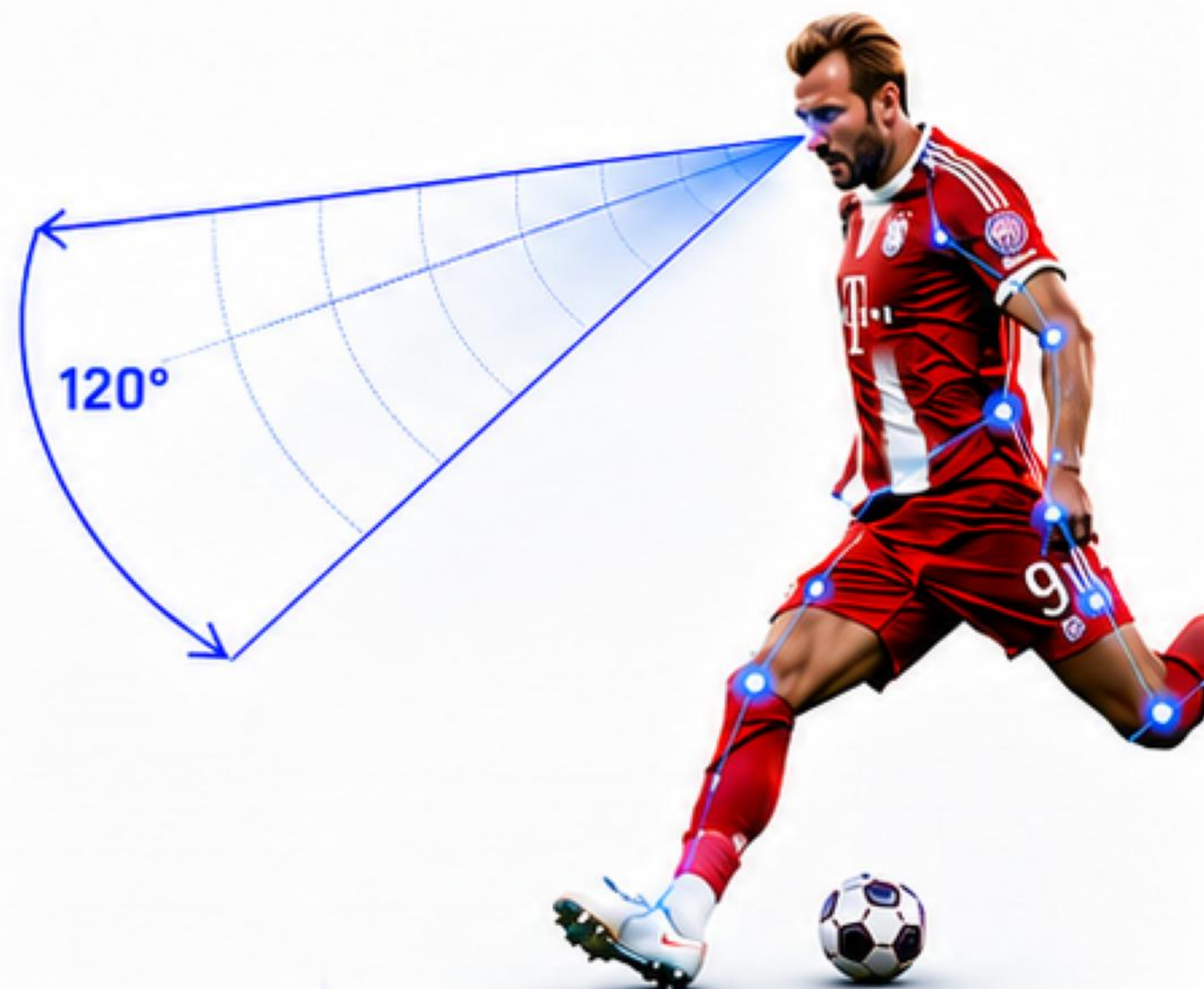
MICHAEL OLISE
FC BAYERN MÜNCHEN



2 • VISION

A 120° cone in front of each player's eyes shows what he can actually see right now.

When a teammate or rival is in the way, the area behind them goes dark.



3 • ORIENTATION-AWARE PITCH CONTROL

A player only controls the ground he can reach without turning his body.

To get behind his own shoulder he has to spin first, so that space is wide open even when he is standing right there.



4 • DOS: DIAGONAL OPPORTUNITY SURFACE

A diagonal pass attacks the side a defender isn't looking.

He spots it too late, can't recover in time, and a clear gap opens up.

DOS shows you exactly where.

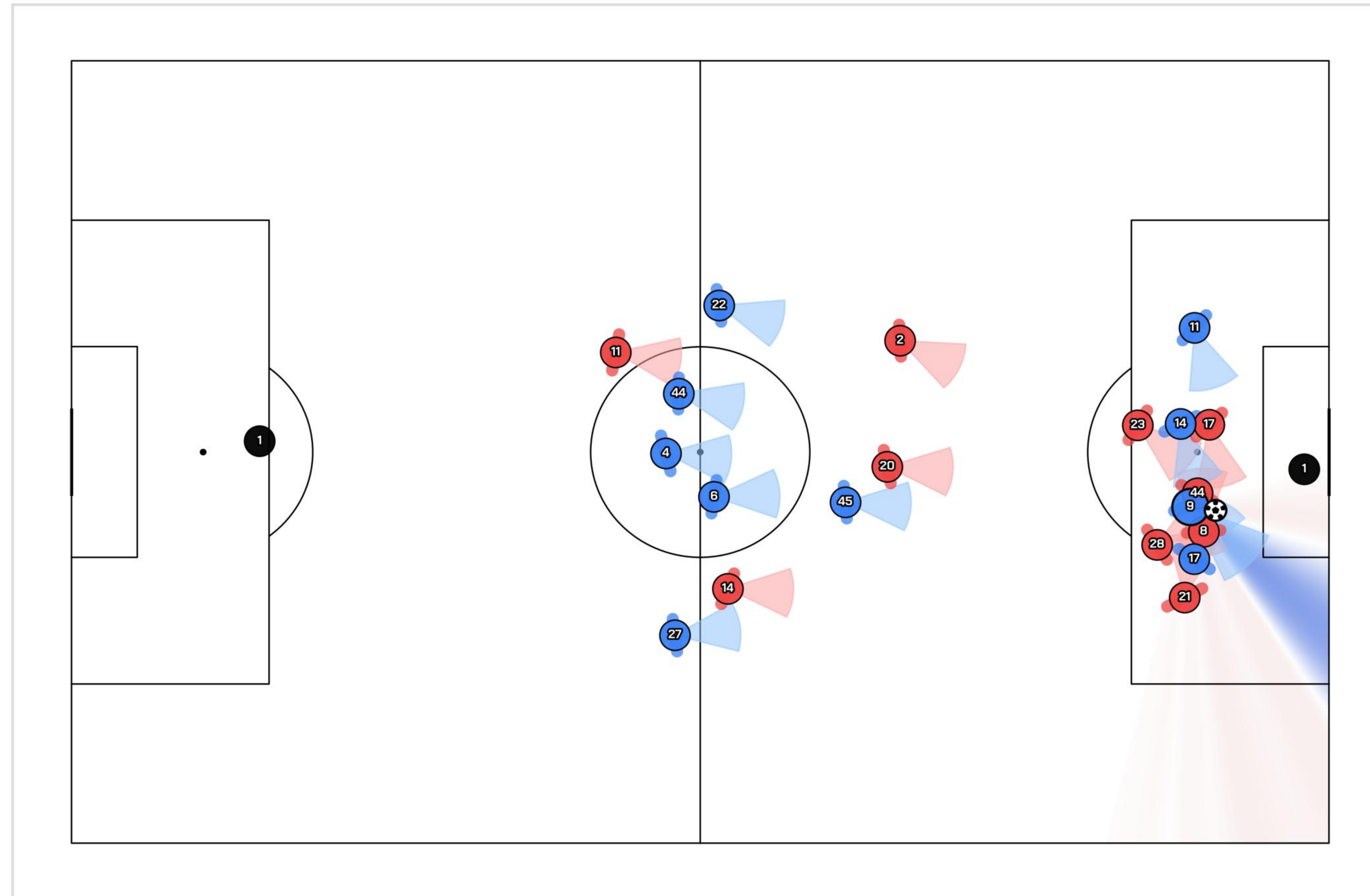


TRACAB 3D SKELETON

• 5 BUNDESLIGA MATCHES 2025-26

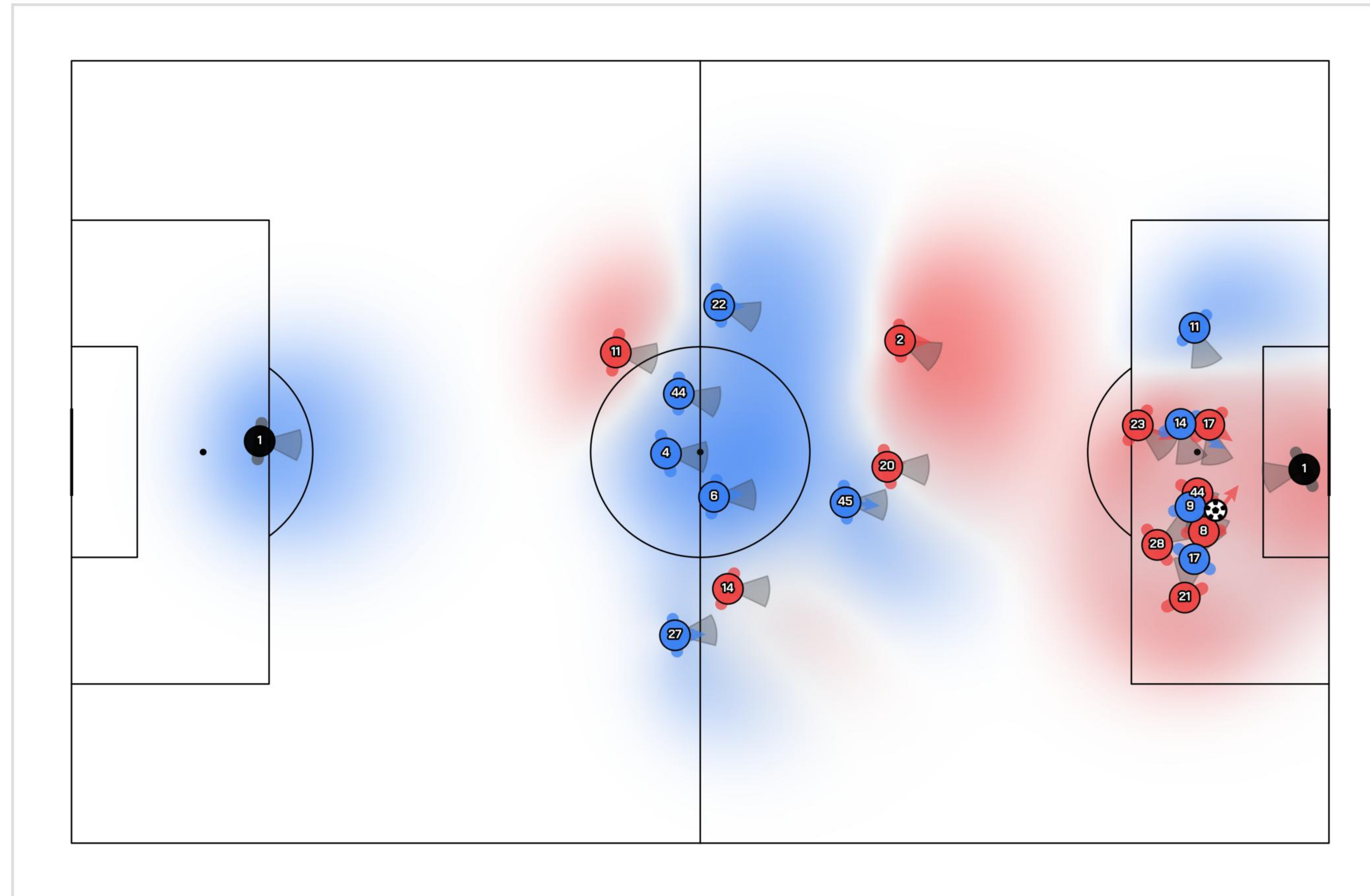
• CREATED BY JAIME ORIOL

What every player can see



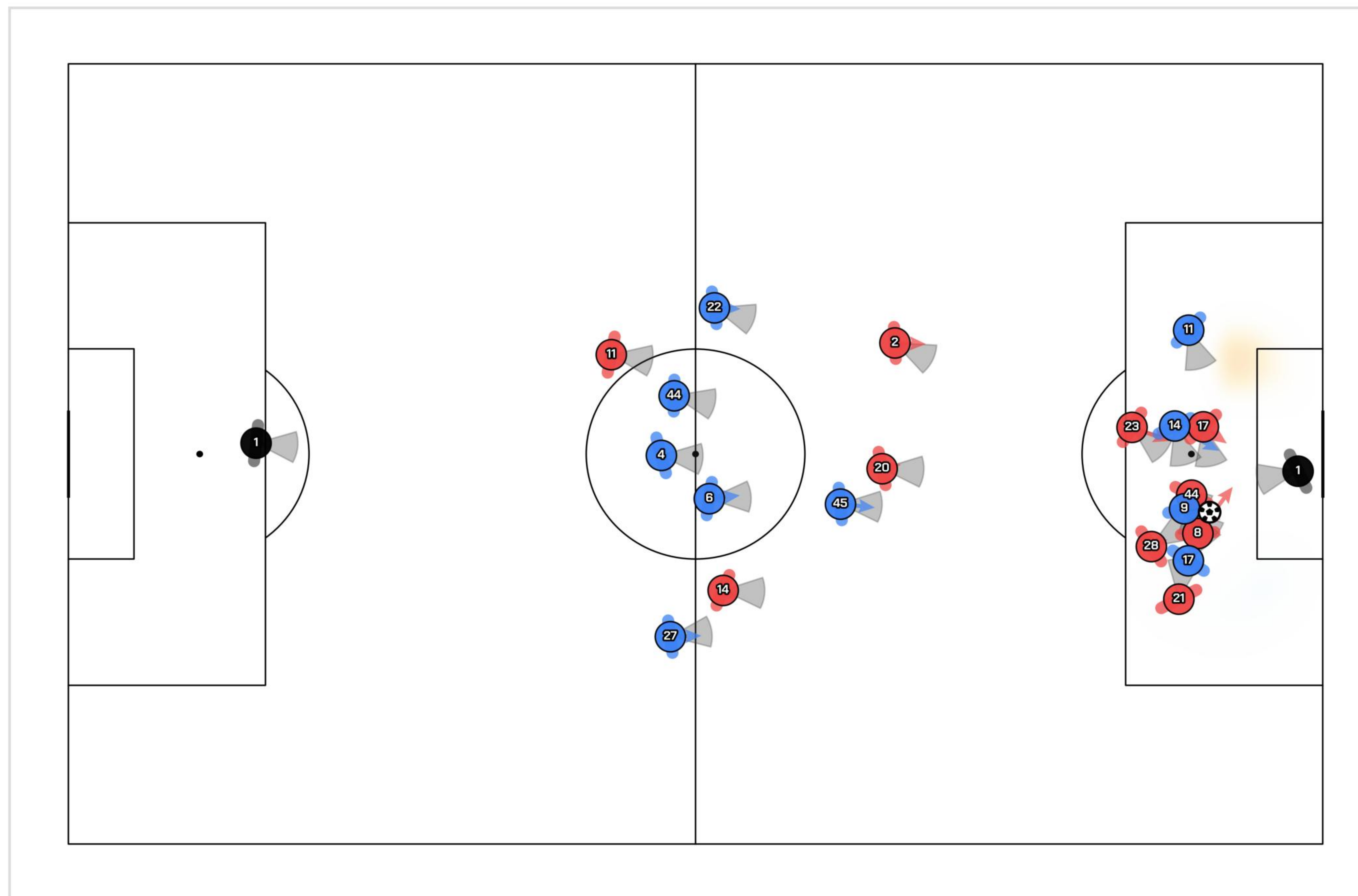
A 120° cone of vision built from where each player is really looking. It narrows when they sprint, and other players block the view.

What every player can reach



How much ground a player covers depends on which way he faces. Behind his shoulder it shrinks, leaving a real gap in the defence.

The Diagonal Opportunity Surface



The extra space a diagonal pass opens up, shown only where the player on the ball can actually see it. Cyan: he sees it. Amber: he lost track of it.